

| <b>(b) Mains electricity</b> |                                                                                                                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Students should:</b>      |                                                                                                                                                                                                                      |
| 2.2                          | understand how the use of insulation, double insulation, earthing, fuses and circuit breakers protects the device or user in a range of domestic appliances                                                          |
| 2.3                          | understand why a current in a resistor results in the electrical transfer of energy and an increase in temperature, and how this can be used in a variety of domestic contexts                                       |
| 2.4                          | <p>know and use the relationship between power, current and voltage:</p> <p>power = current <math>\times</math> voltage</p> $P = I \times V$ <p>and apply the relationship to the selection of appropriate fuses</p> |
| 2.5                          | <p>use the relationship between energy transferred, current, voltage and time:</p> <p>energy transferred = current <math>\times</math> voltage <math>\times</math> time</p> $E = I \times V \times t$                |
| 2.6                          | know the difference between mains electricity being alternating current (a.c.) and direct current (d.c.) being supplied by a cell or battery                                                                         |